WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS2 31/03/2026

Beatriz Francisco Nº 118638
Catarina Ribeiro Nº 119467
Mariana Marques Nº118971
Marta Cruz Nº 119572
Matilde Rodrigues Nº119714

Projeto em Engenharia Informática 2025/2026

Motion Suit

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies

Related Work

	SmartPosture	FlexiTrace	SwordHealth	DorsaVI	Current Suit	MotionSuit
Performance Analytics	✓	✓	✓	✓	✓	✓
Real-Time Feedback	✗	✗	✓	✓	✓	✓
Wearable Sensors	✗	✗	✗	✓	✓	✓
Gamification	✗	✓	✗	✗	✗	✓

Personas

	As a	I want	So that
	Suit User	View statistics on dashboard	I can see my posture data over time
	Researcher	Access all collected information	I can identify postural deviations and abnormal patterns
	Administrator	To manage user accounts	I can control acess and provide support

Functional

SUIT USER

01 The system must allow to register and log in with their respective role

02 The system must deliver posture-related alerts to users

03 The system must provide a dashboard displaying all personal user data

04 The system must enable users to export their personal data

Functional

RESEARCHER

05 The system must display all user data with multiple views and selectable time periods.

06 The system must allow researchers to export data from all users.

Functional

ADMINISTRATOR

07 The system must provide user account management capabilities

08 The system must provide an administrative dashboard displaying system metrics.

Non-Functional

Security and Privacy

01 The system must encrypt all the Protected Health Information

02 The system must store passwords using salted cryptographic hashes

Reliability

Usability and User Experience

Data Management

Maintainability and Scalability

Non-Functional

Security and Privacy

Reliability

03 The system must guarantee zero data loss at the end of each session

04 The system must support offline operability through local data caching

Usability and User Experience

Data Management

Maintainability and Scalability

Non-Functional

Security and Privacy

Reliability

Usability and User Experience

05 The system must allow new users to perform basic monitoring tasks

06 The system must enable users to personalize dashboard layouts

Data Management

Maintainability and Scalability

Non-Functional

Security and Privacy

Reliability

Usability and User Experience

Data Management

07 The system must ensure high-performance queries for user-facing operations

08 The system must provide reliable retention of raw sensor data

Maintainability and Scalability

Non-Functional

Security and Privacy

Reliability

Usability and User Experience

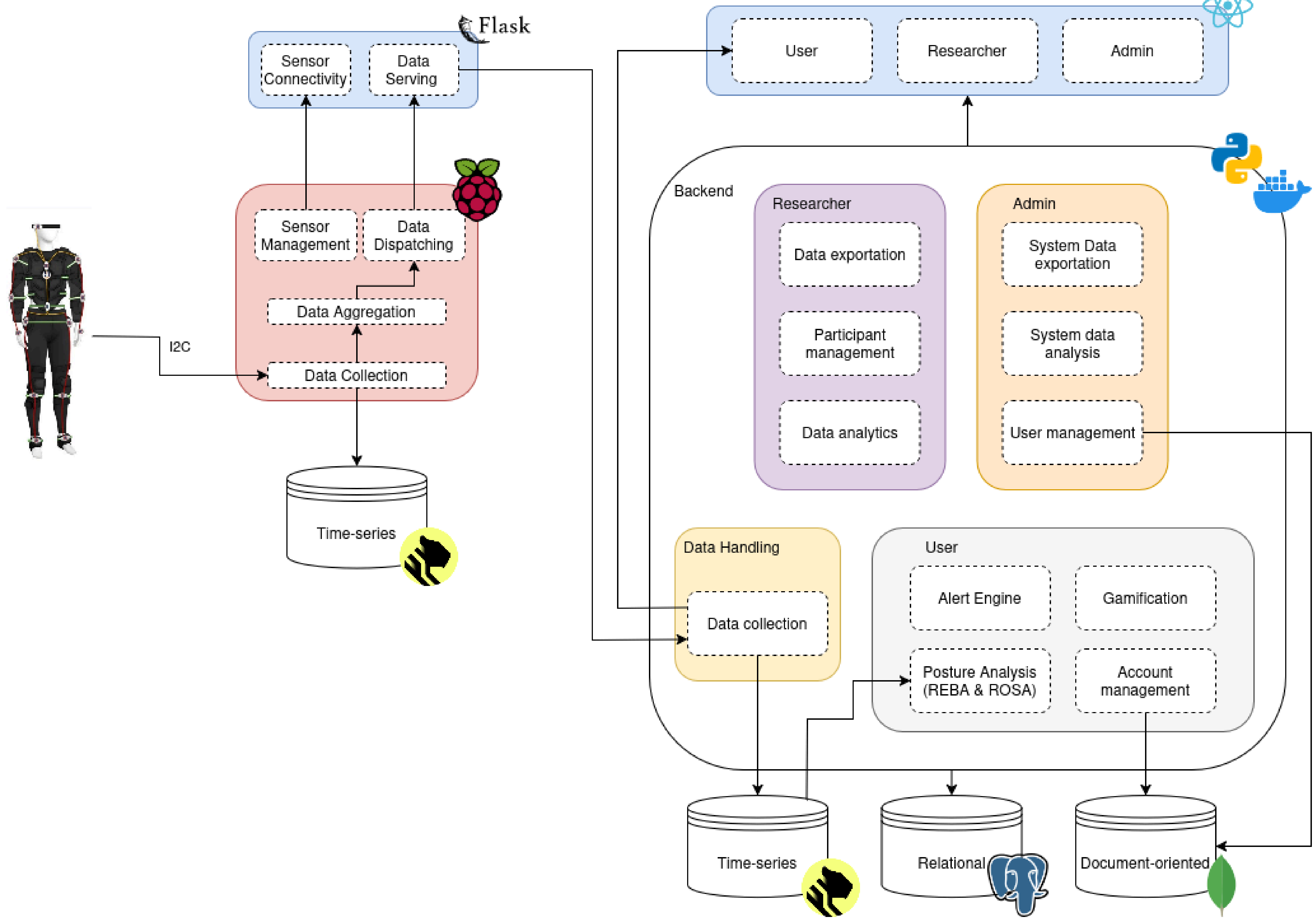
Data Management

Maintainability and Scalability

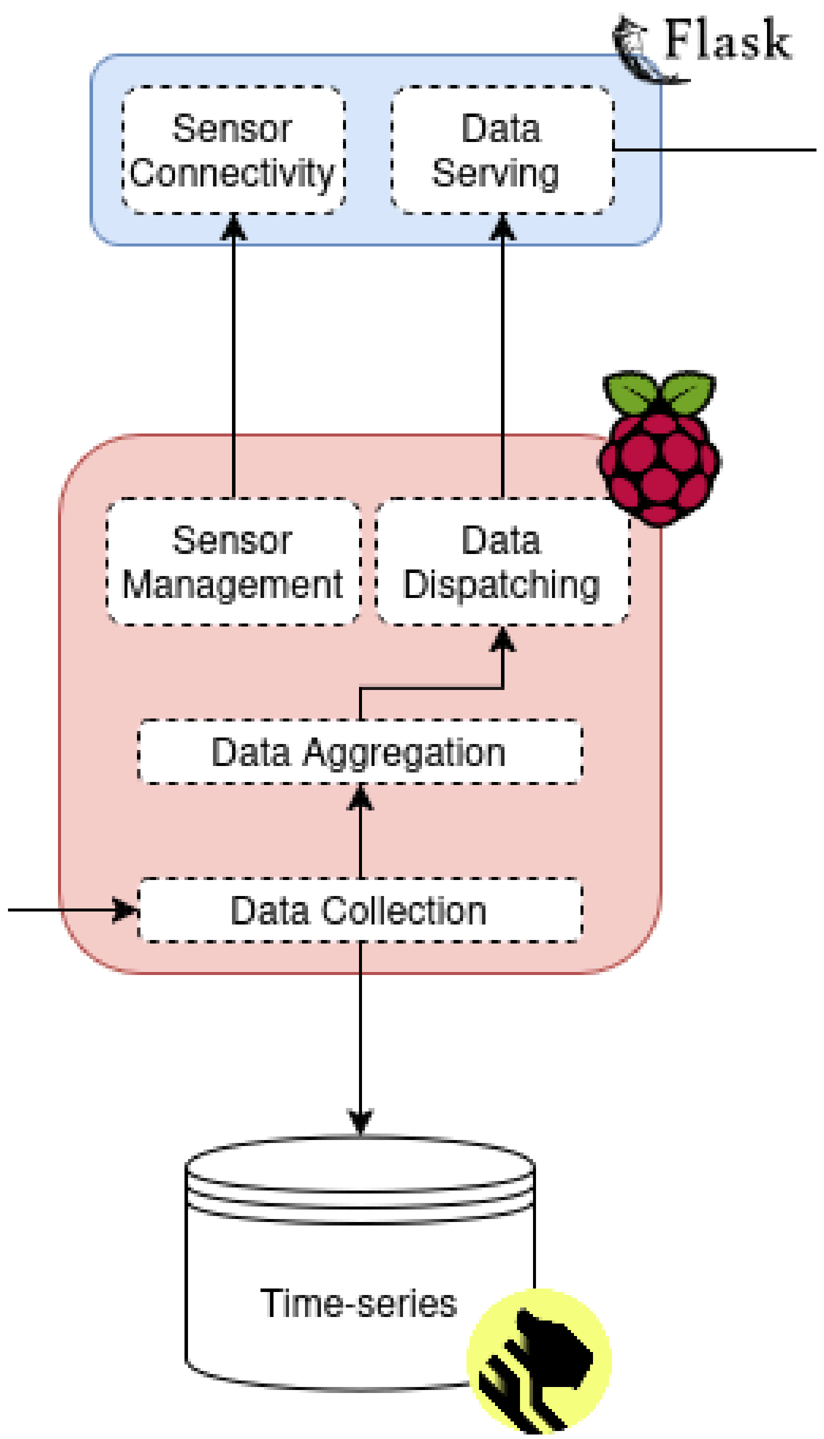
09 The system must expose well-defined APIs for all components

10 The system must support the addition of new sensor types through its sensor integration

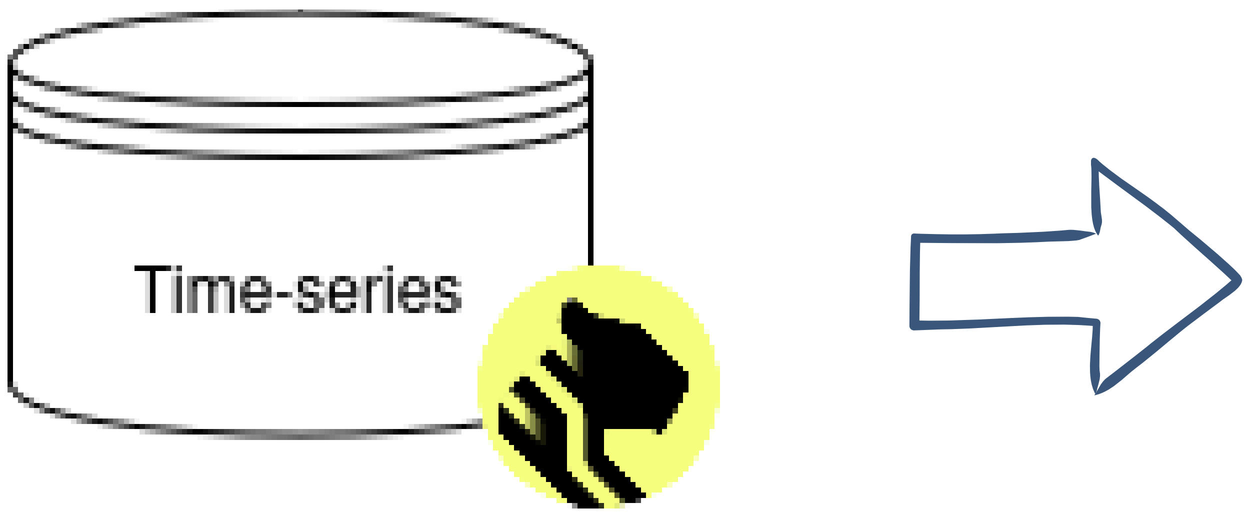
Architecture



Architecture

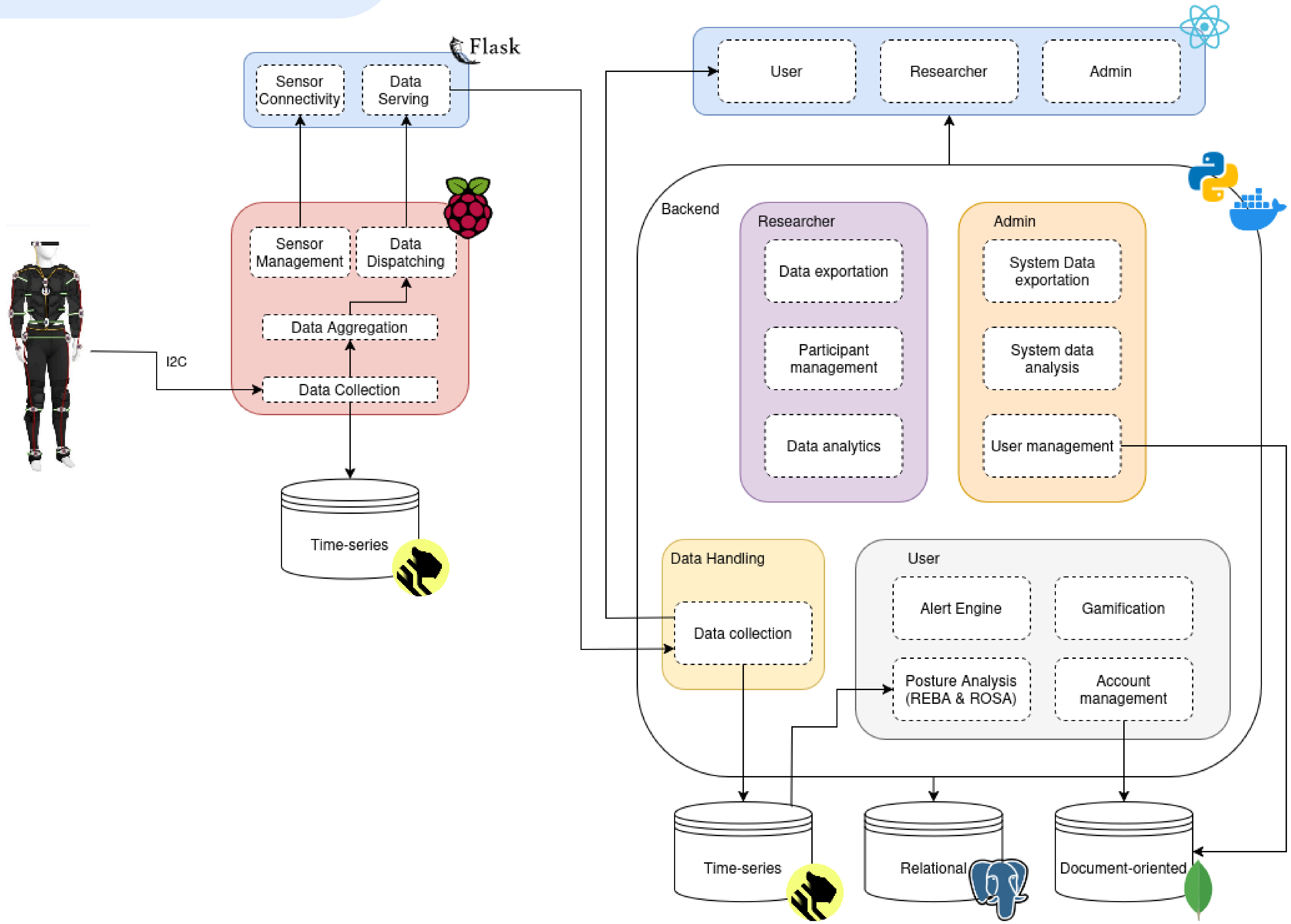


Architecture

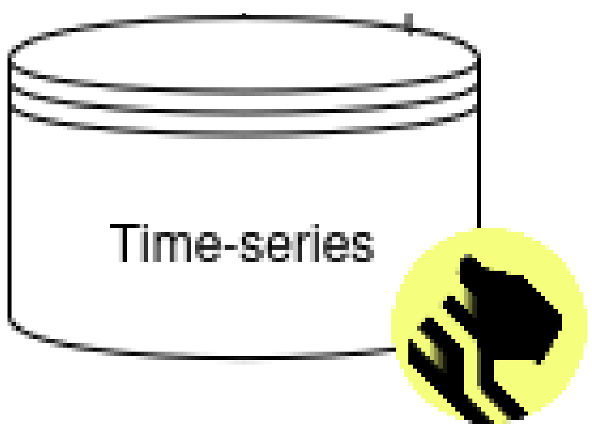


Sensor Data
SensorID
Location_On_Body
Session_ID
User_ID
Frame_Number
Temperature
Posture_Label
Quaternion
Acceleration
Gyroscopoe
Euler
Calibration
Neck_Pitch
Upper_Arm_Pitch
Lower_Arm_Pitch
Knee_Pitch

Architecture



Architecture

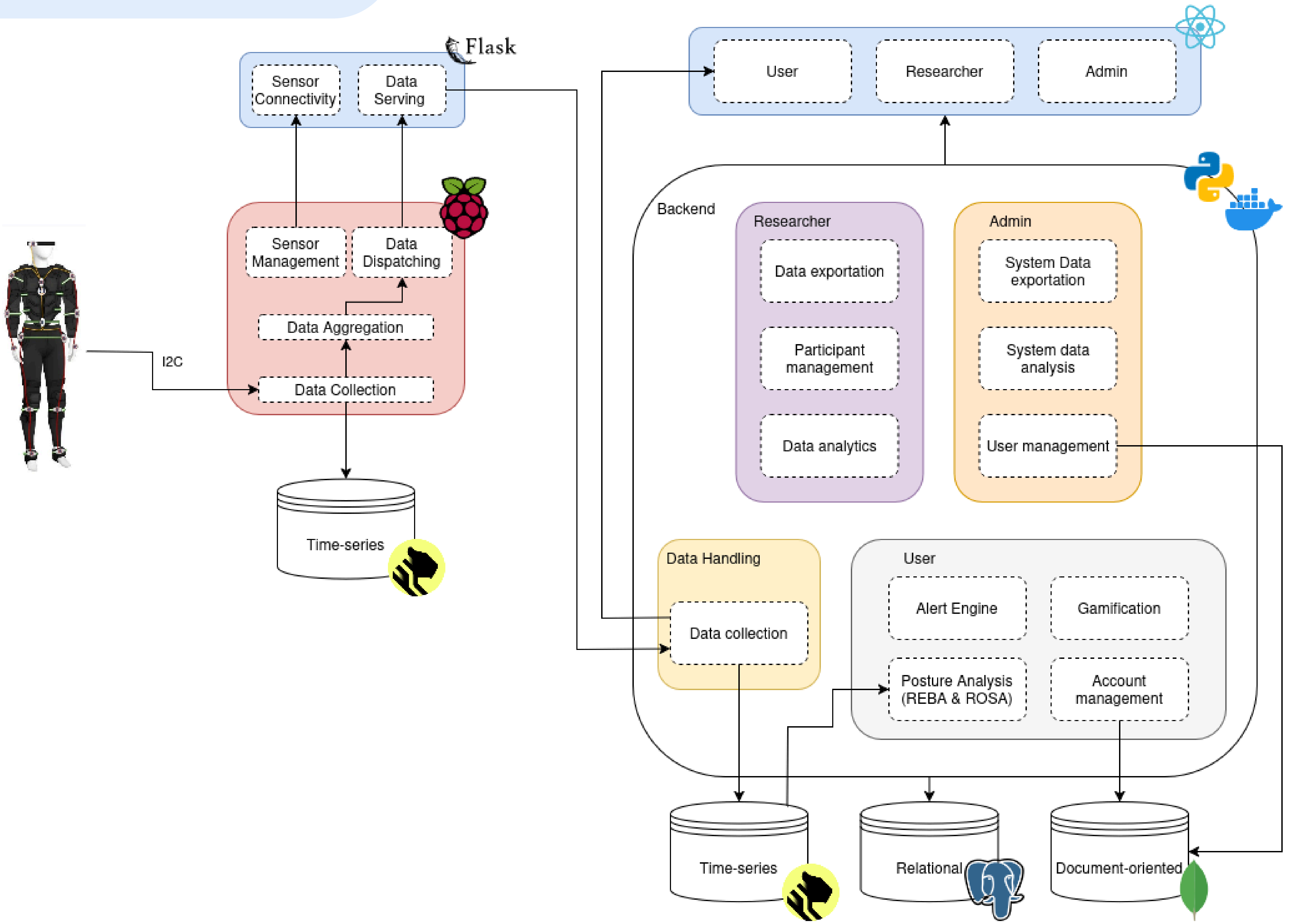


Sensor Data
Sensor_ID
Location_On_Body
Session_ID
User_ID
Frame_Number
Temperature
Posture_Label
Quaternion
Acceleration
Gyroscopoe
Euler
Calibration

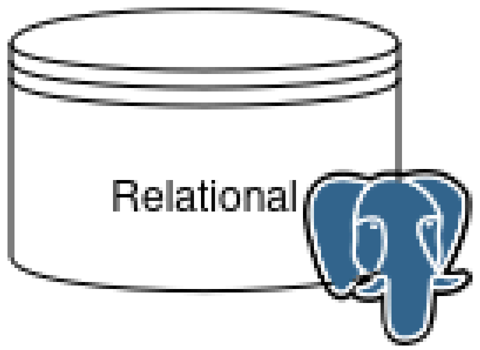
Session
Session_ID
Timestamp_Start
User_ID
Session_Status
Timestamp_Start
Timestamp_End
Duration_Seconds
Avg_Posture
Max_Posture
Min_Posture
Total_Alerts
Critical_Alerts
Sensor_Locations
Processed_At

Posture Alert
Alert_ID
Timestamp_Start
Session_ID
Alert_Type
Severity
Joint_Angles
Affected_Joints
Sensor_Data_Time
Sensor_Location
Alert_Duration
Is_Resolved

Architecture



Architecture



Admin
UserID

AlertRule
RuleID
UserID
RuleType
RuleName
AlertLevel
AlertMessage

User
UserID
FirstName
LastName
Email
Password_Hash
Birth_Date
Gender
Role
Is_Active
Accepted_Terms
Last_Login_At

Researcher
UserID
Institution
Approval_Status

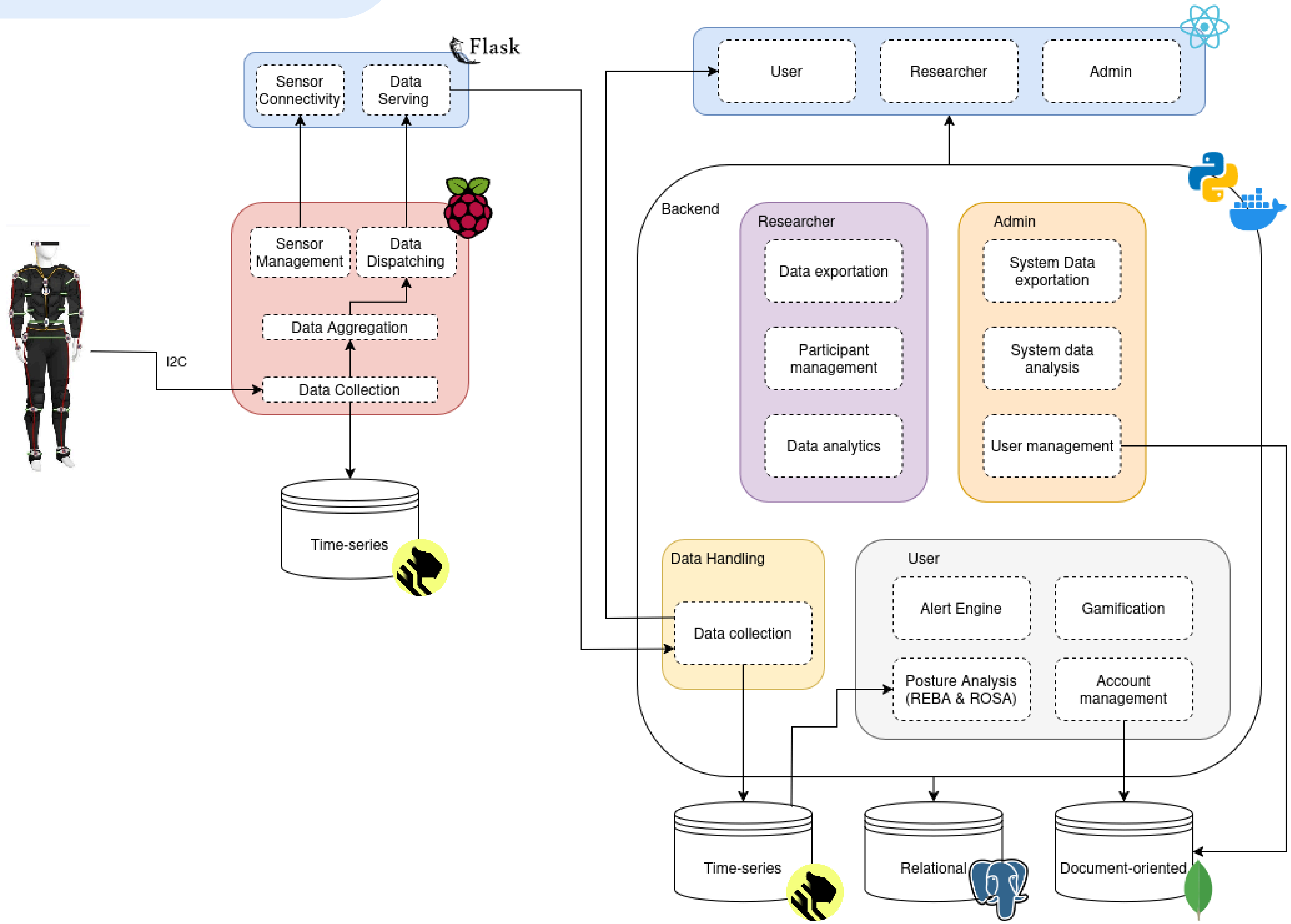
Suit User
UserID
Weight
Height
Occupation
Injury_History
Is_Colorblind
Tts_Enabled

Gamification
UserID
XP_Points
Level
Longest_Streak
Current_Streak

User Preferences
PreferenceID
RealTimeAlerts
Notifications
UserID

Goal
Goal_ID
Goal_Type
Title
Deadline
Target_Value

Architecture

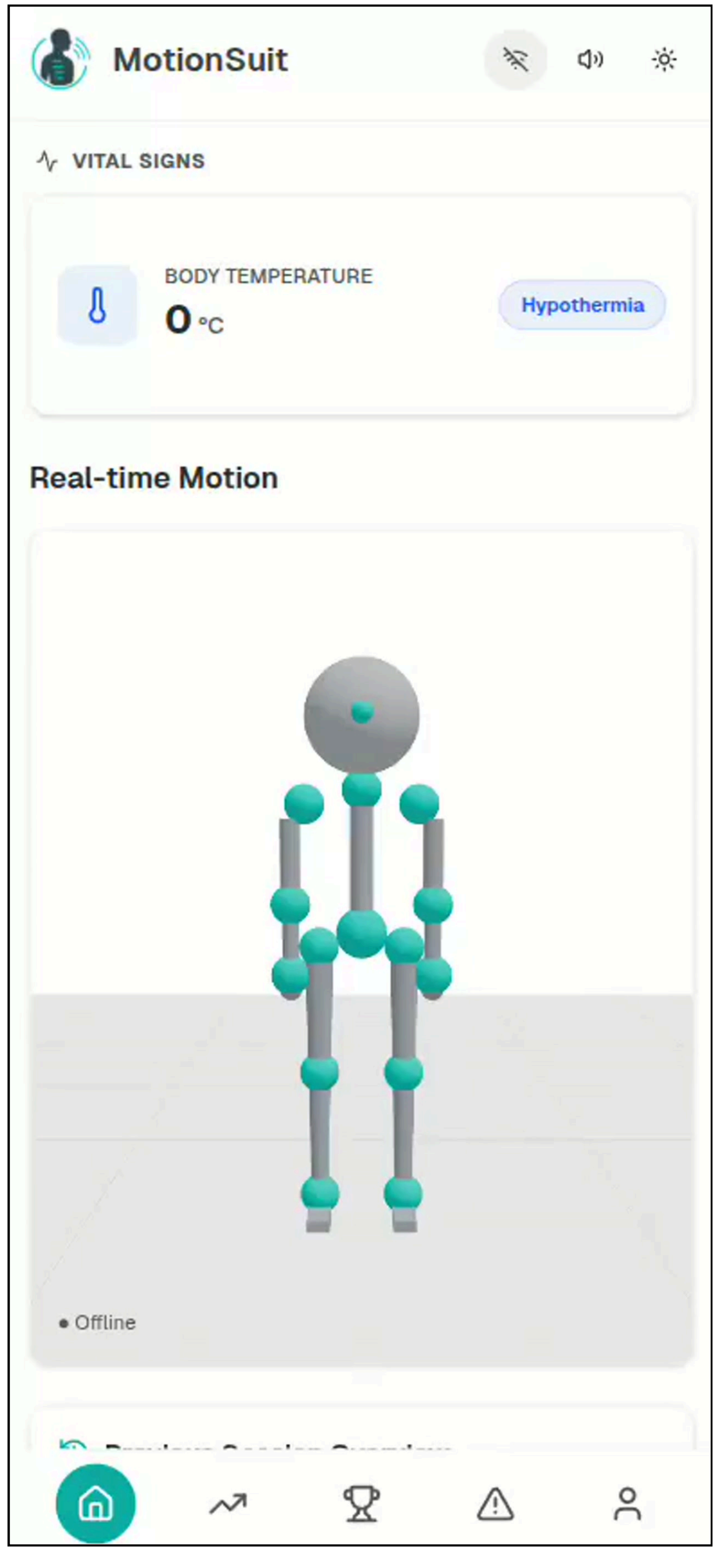


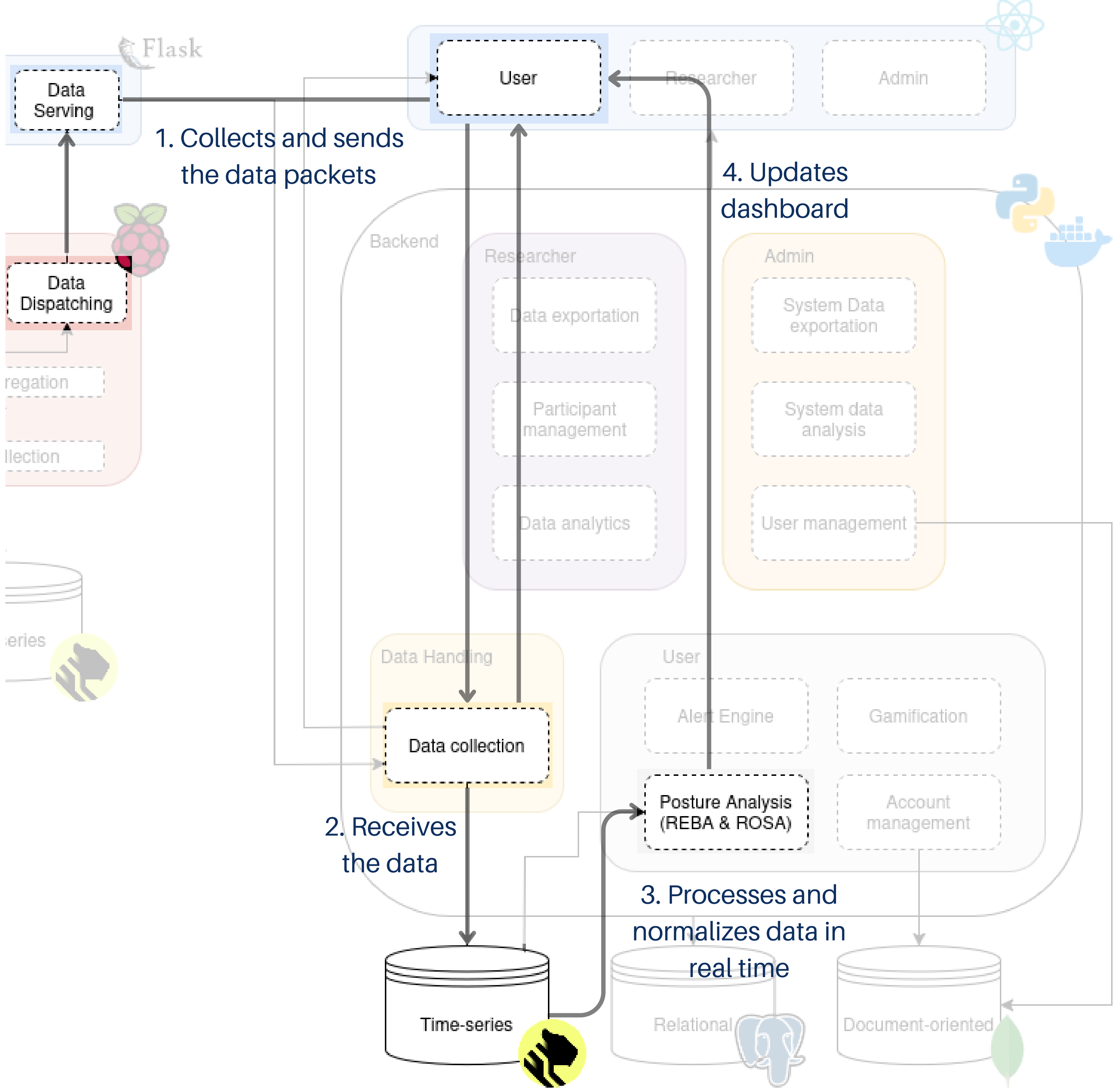
Architecture

User Baseline
BaselineID
UserID
NeckAngleLimit
BackAngleLimit
ShoulderAngleLimit
KneeAngleLimit
LastCalculatedAt
RiskTolerance
BaselineStatus

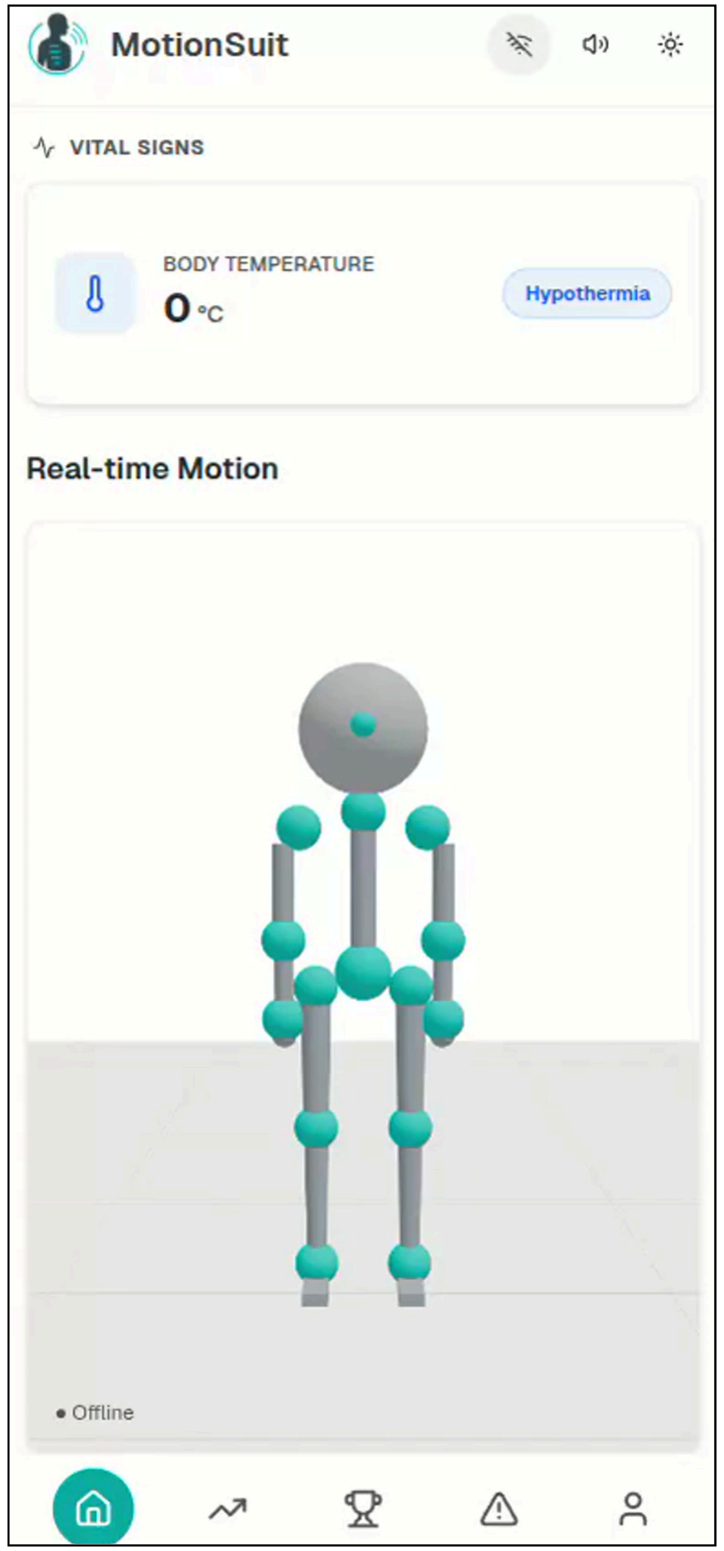
Monitor Real-Time Metrics

REBA Analysis



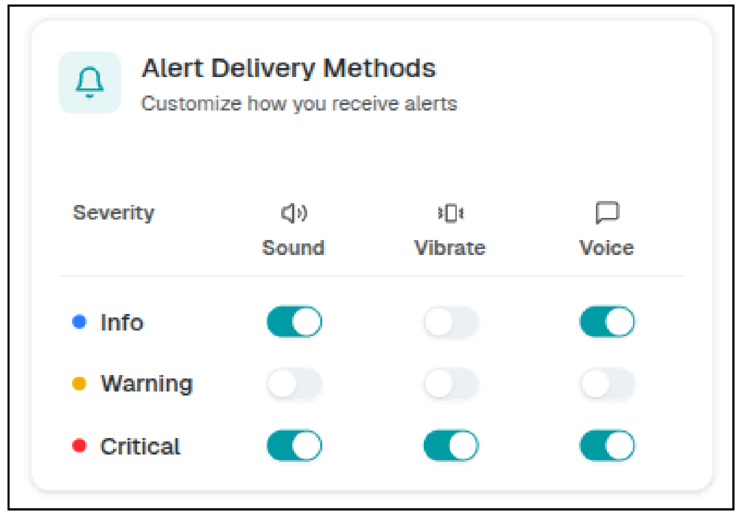


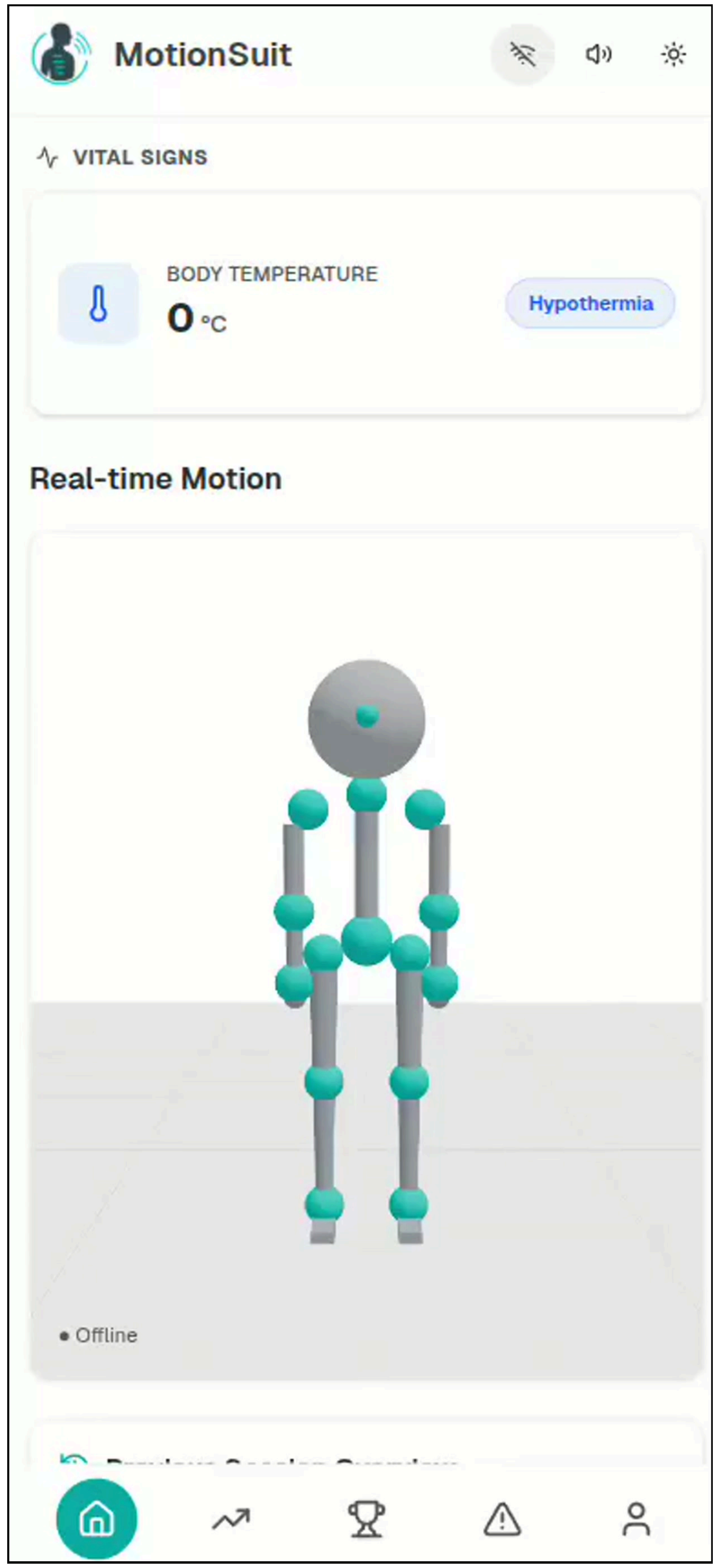
ROSA Analysis

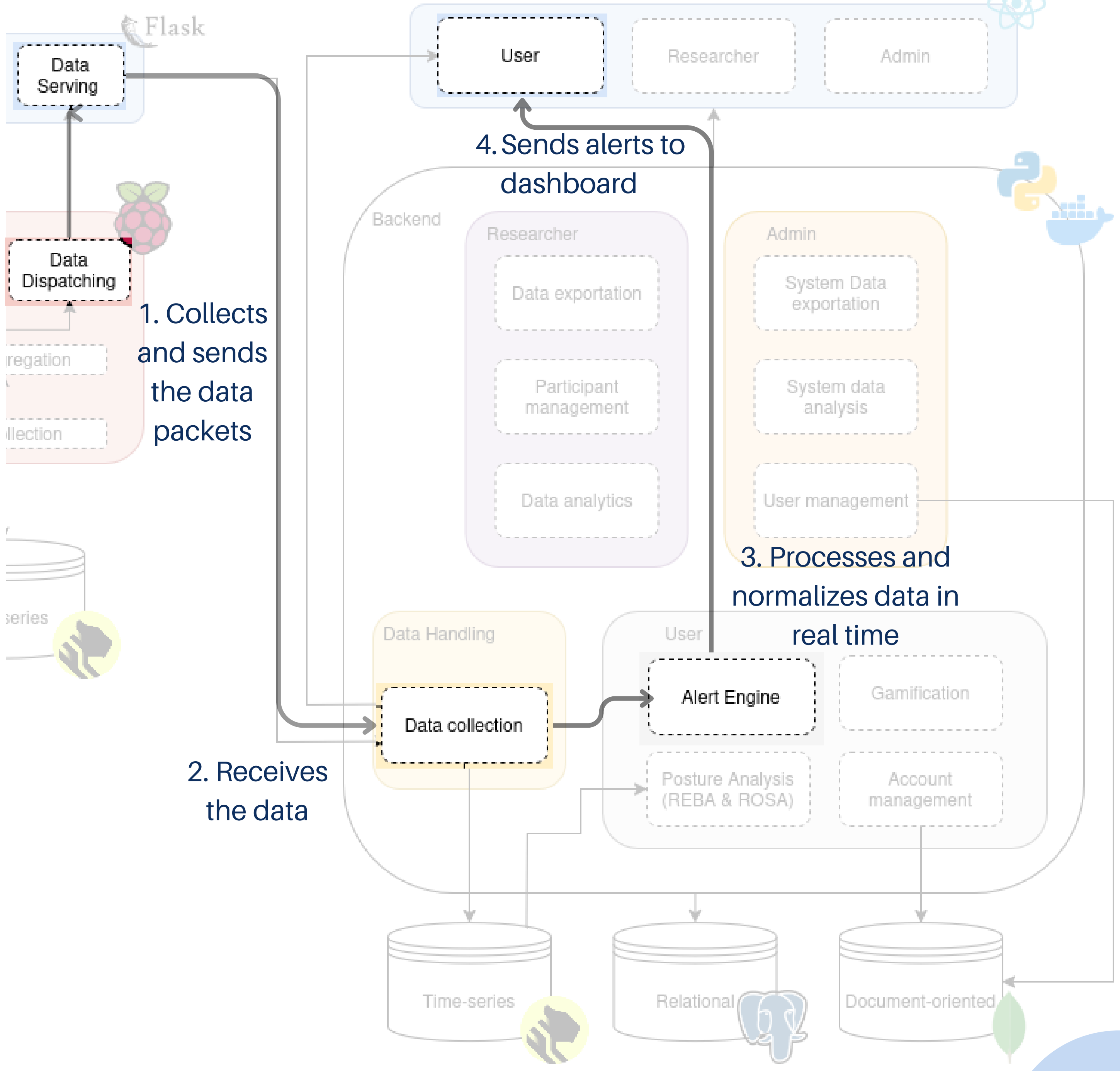


Receive Posture Alerts

Alerts Configuration

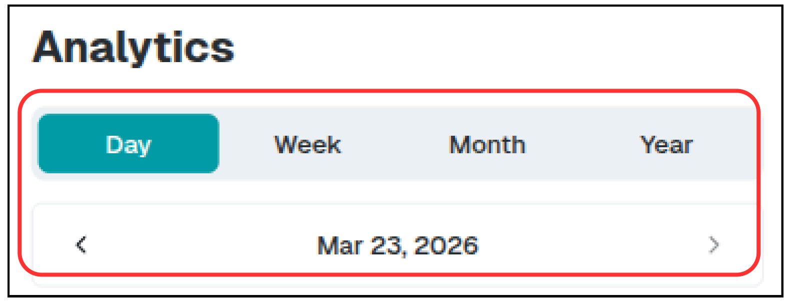


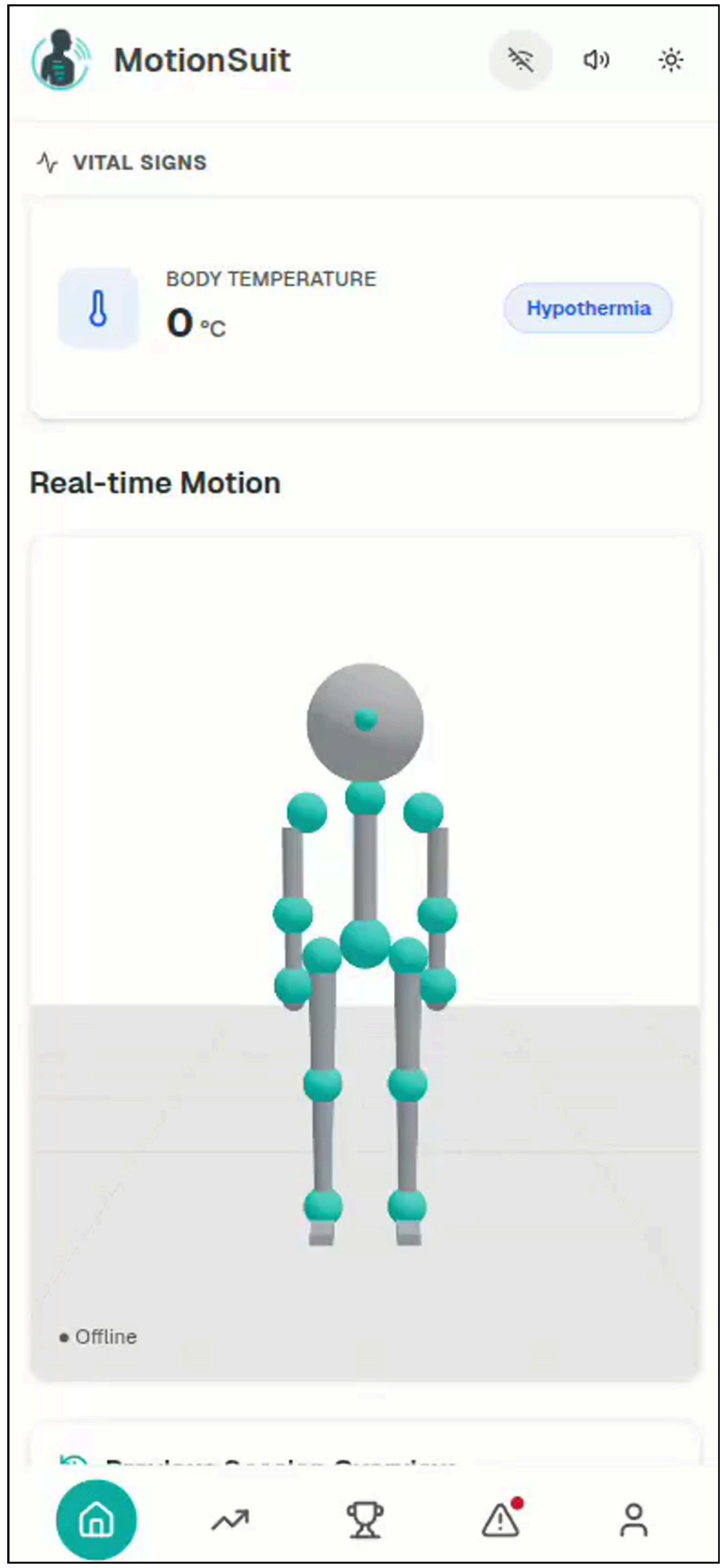


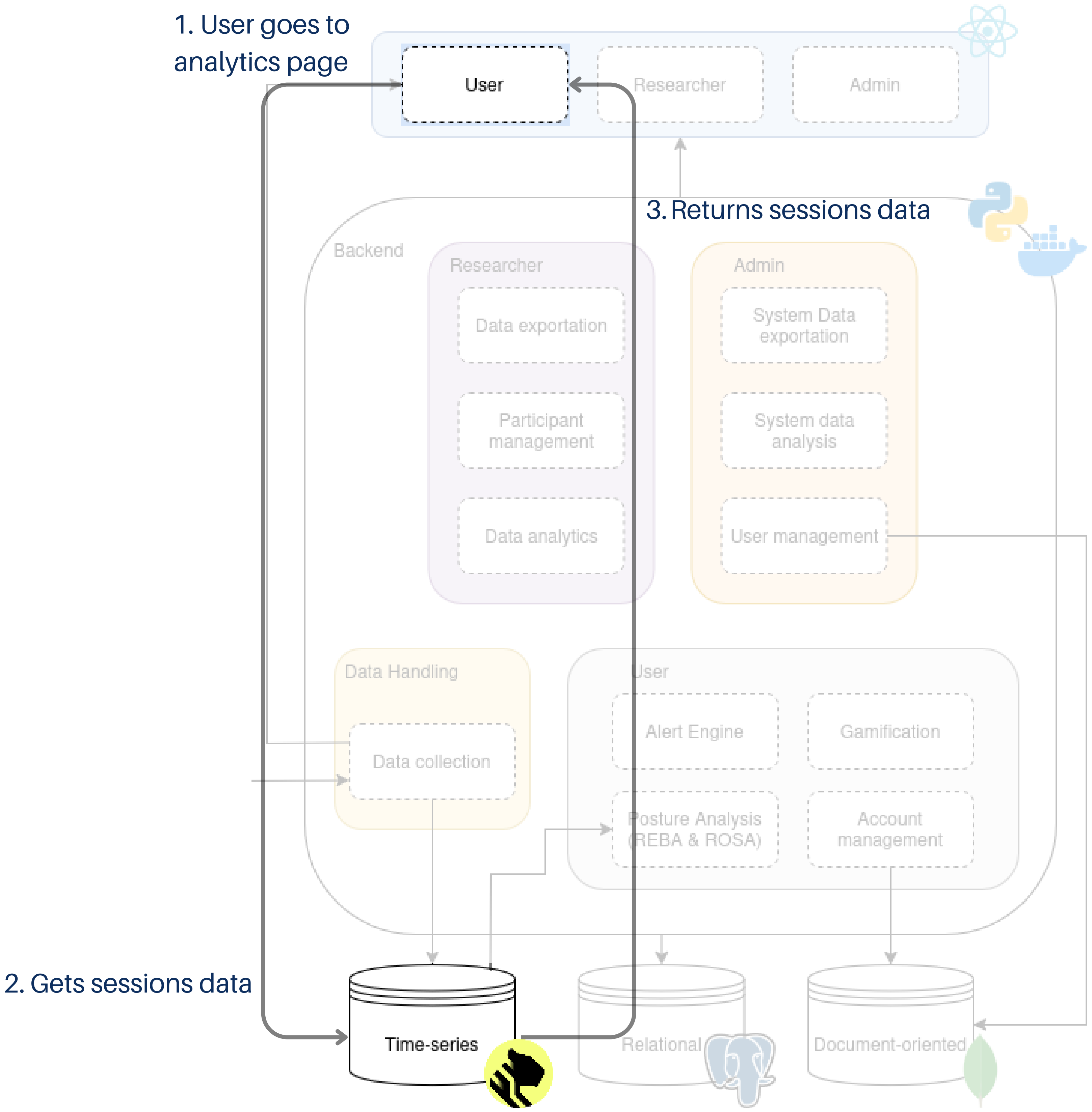


View Historical Statistics

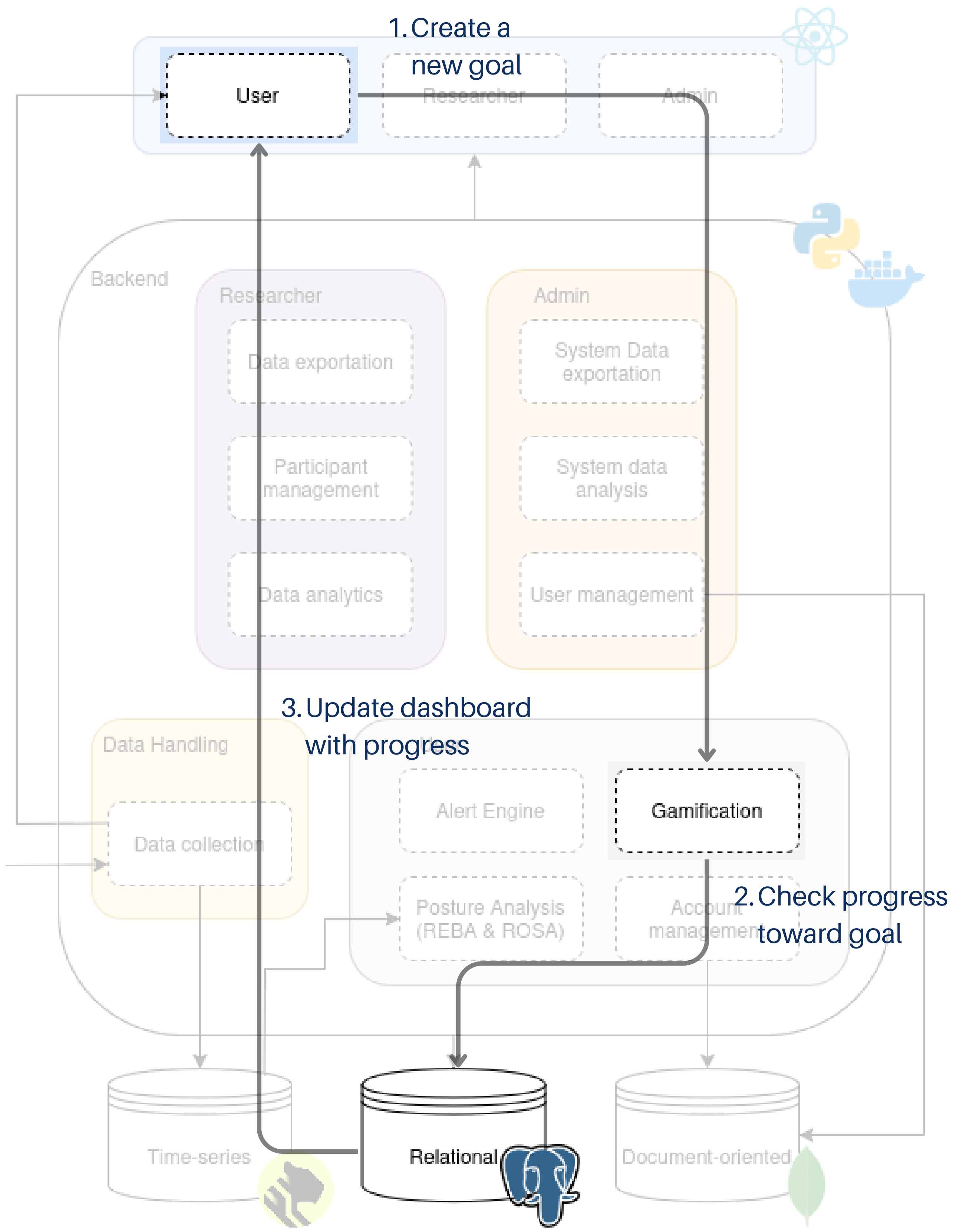
User can choose the time he wants to see his sessions

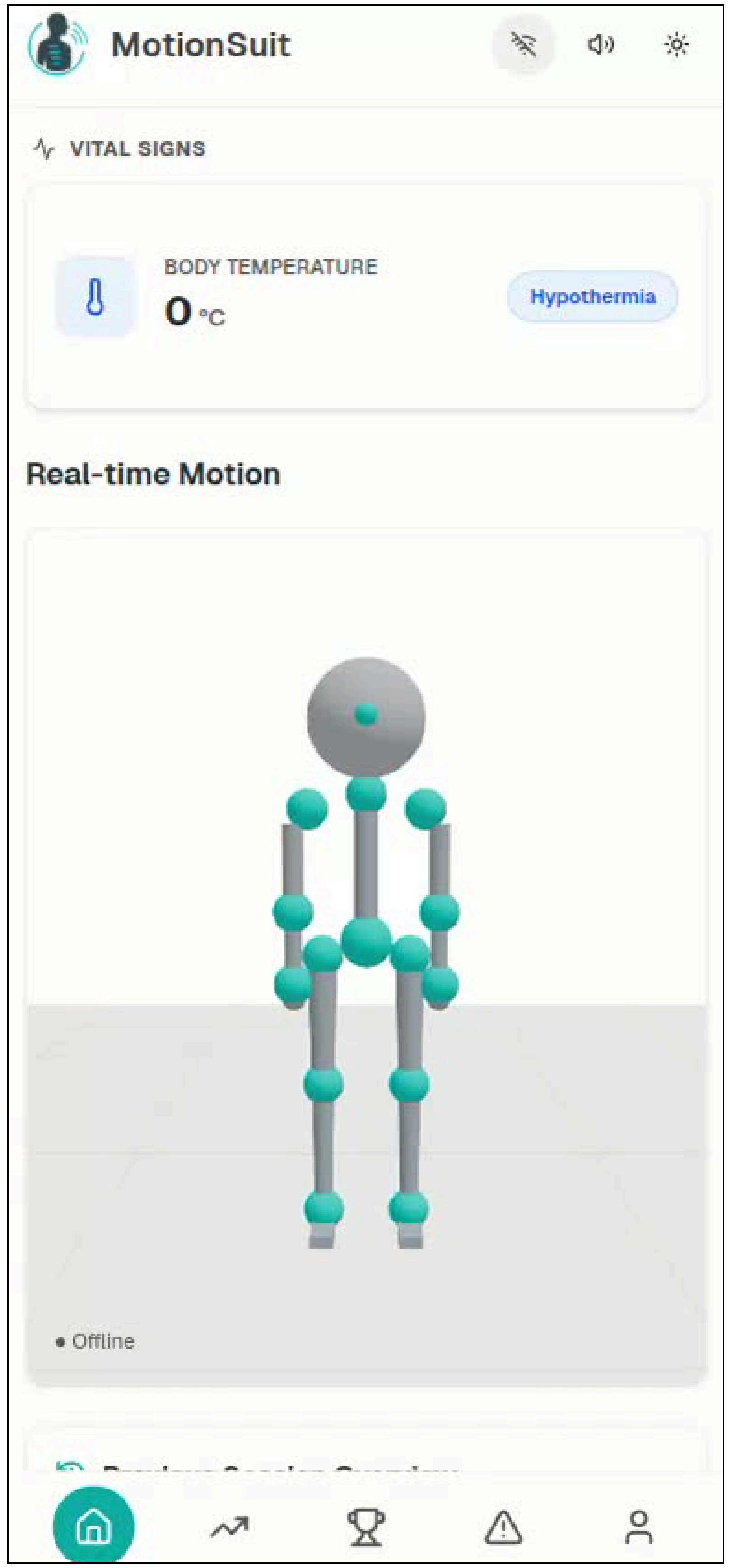




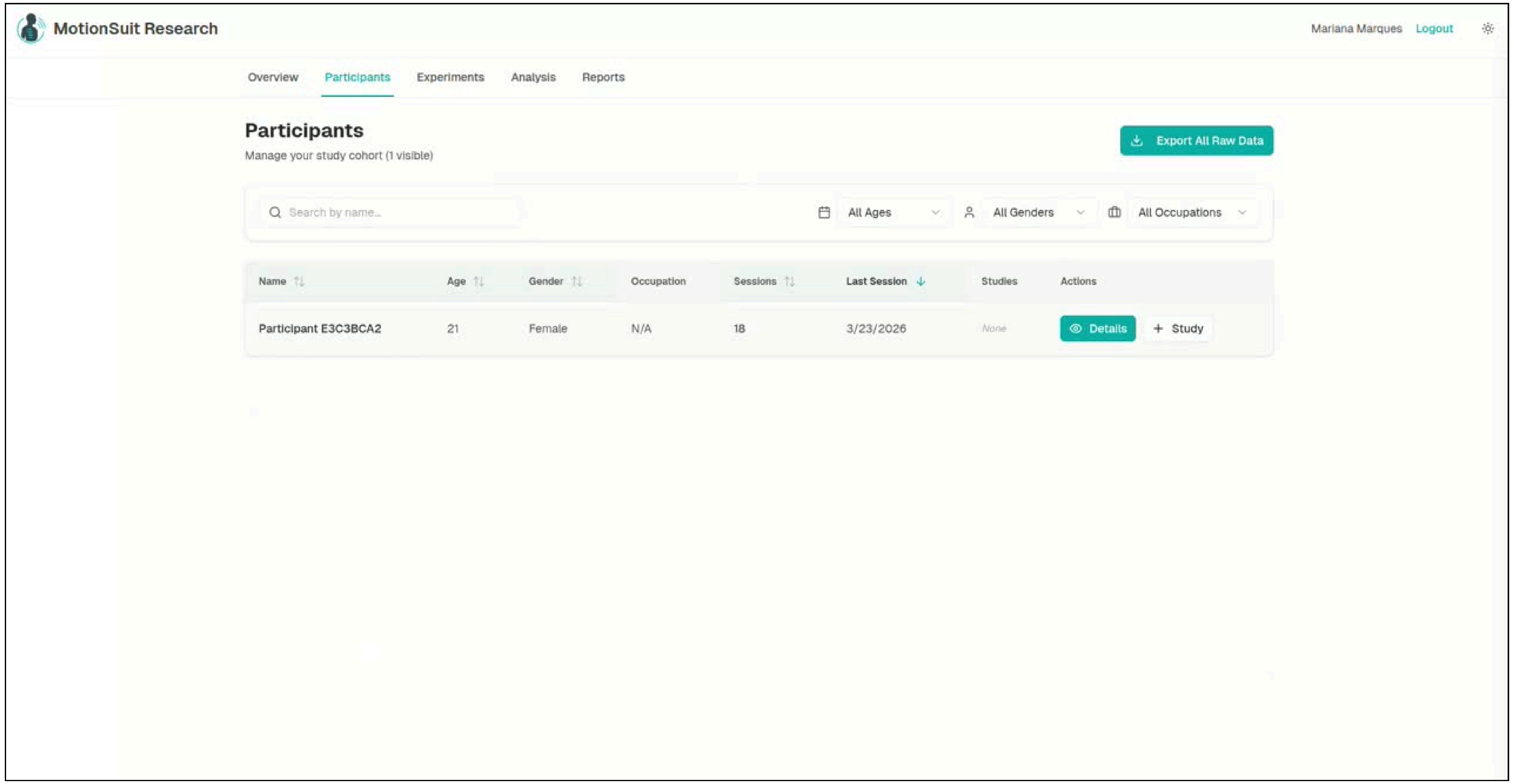


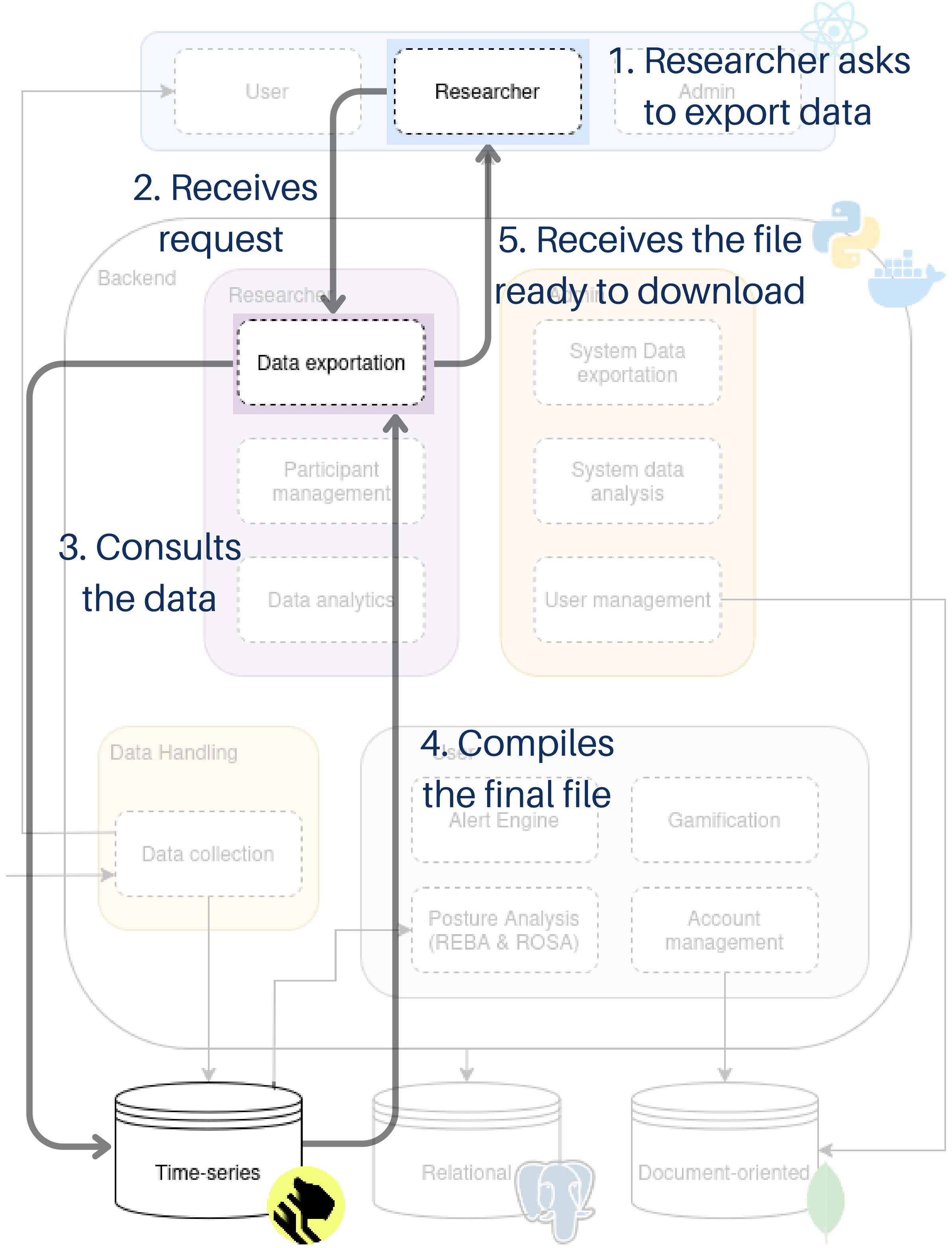
Define Improvement Goals



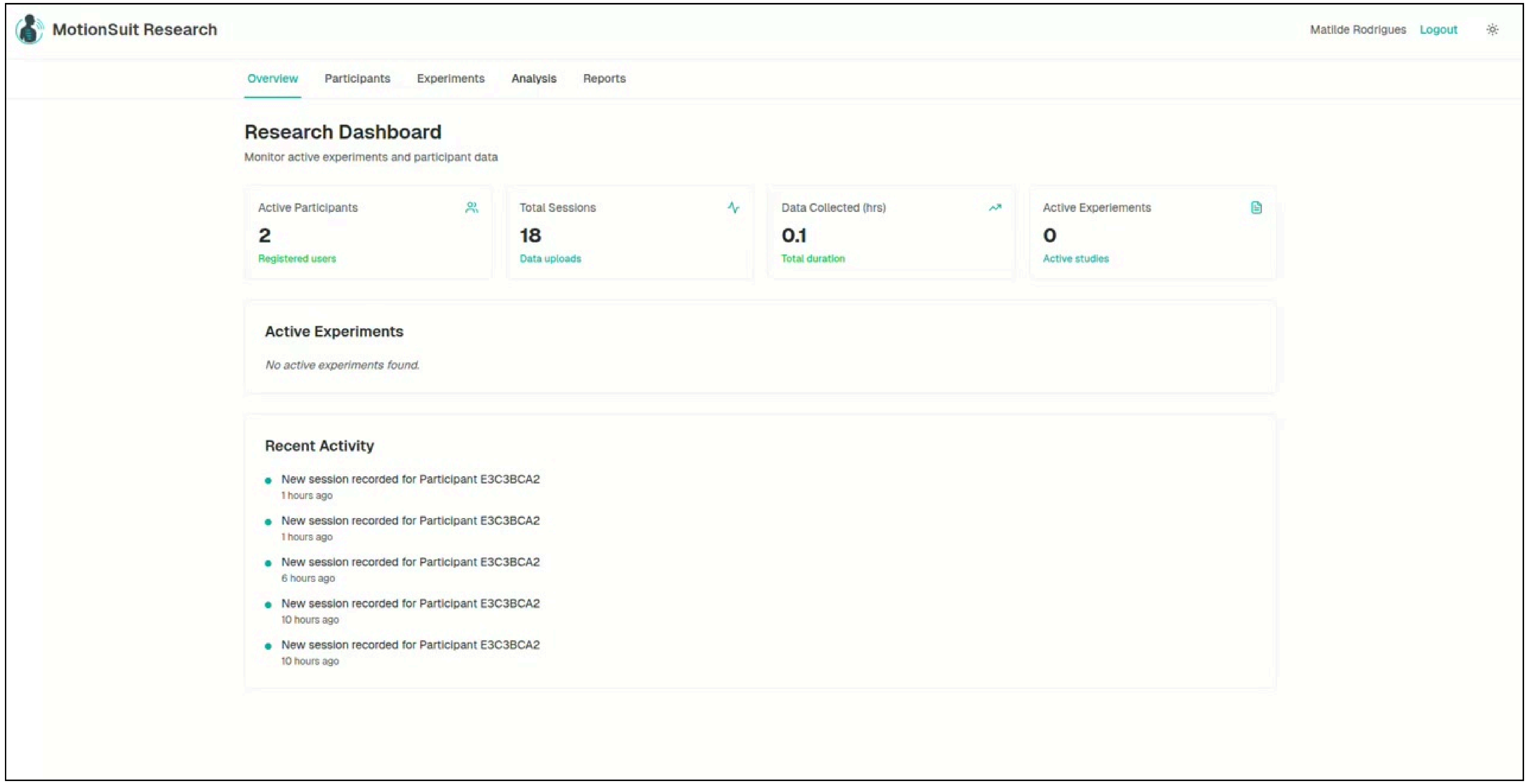


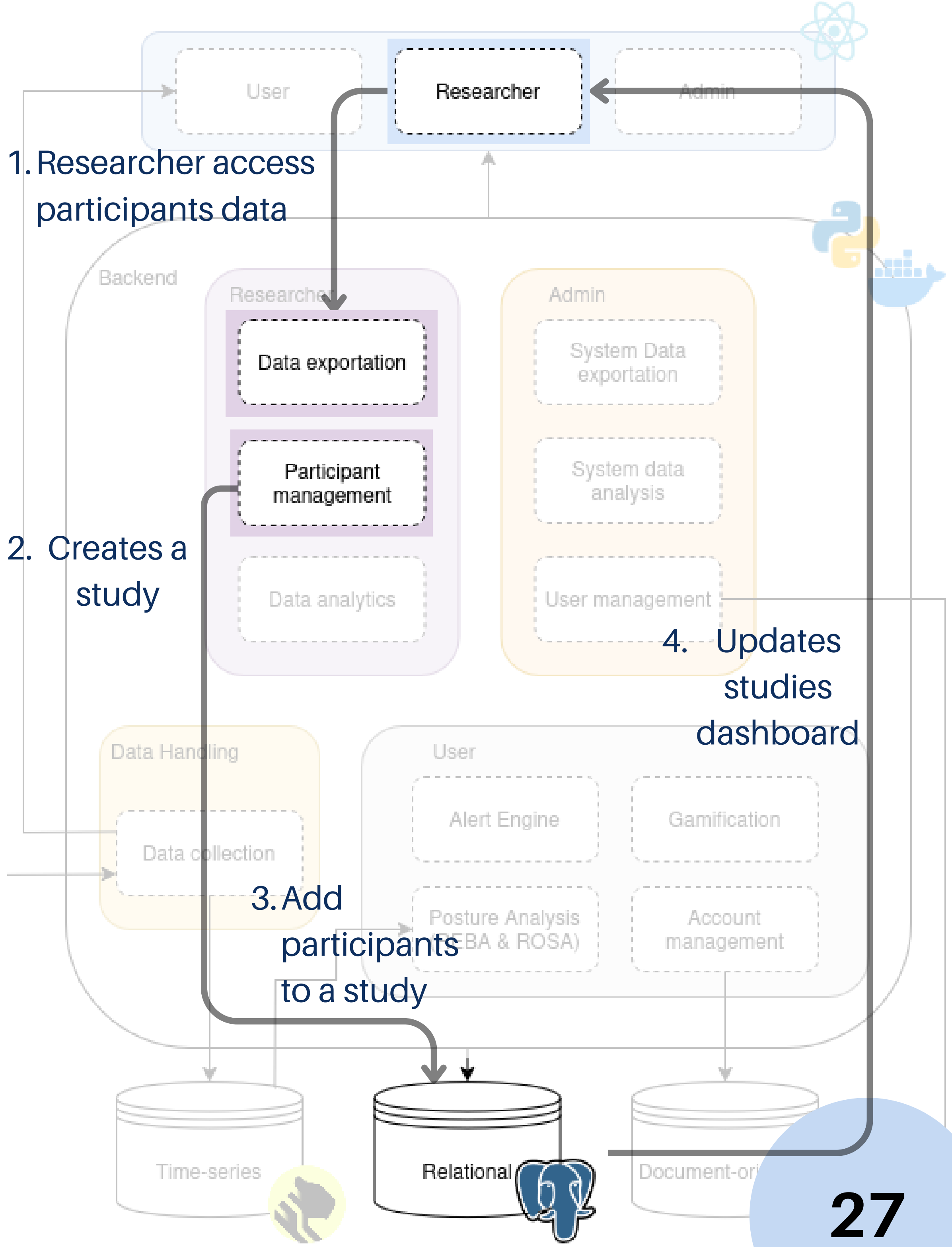
Access All Collected Data



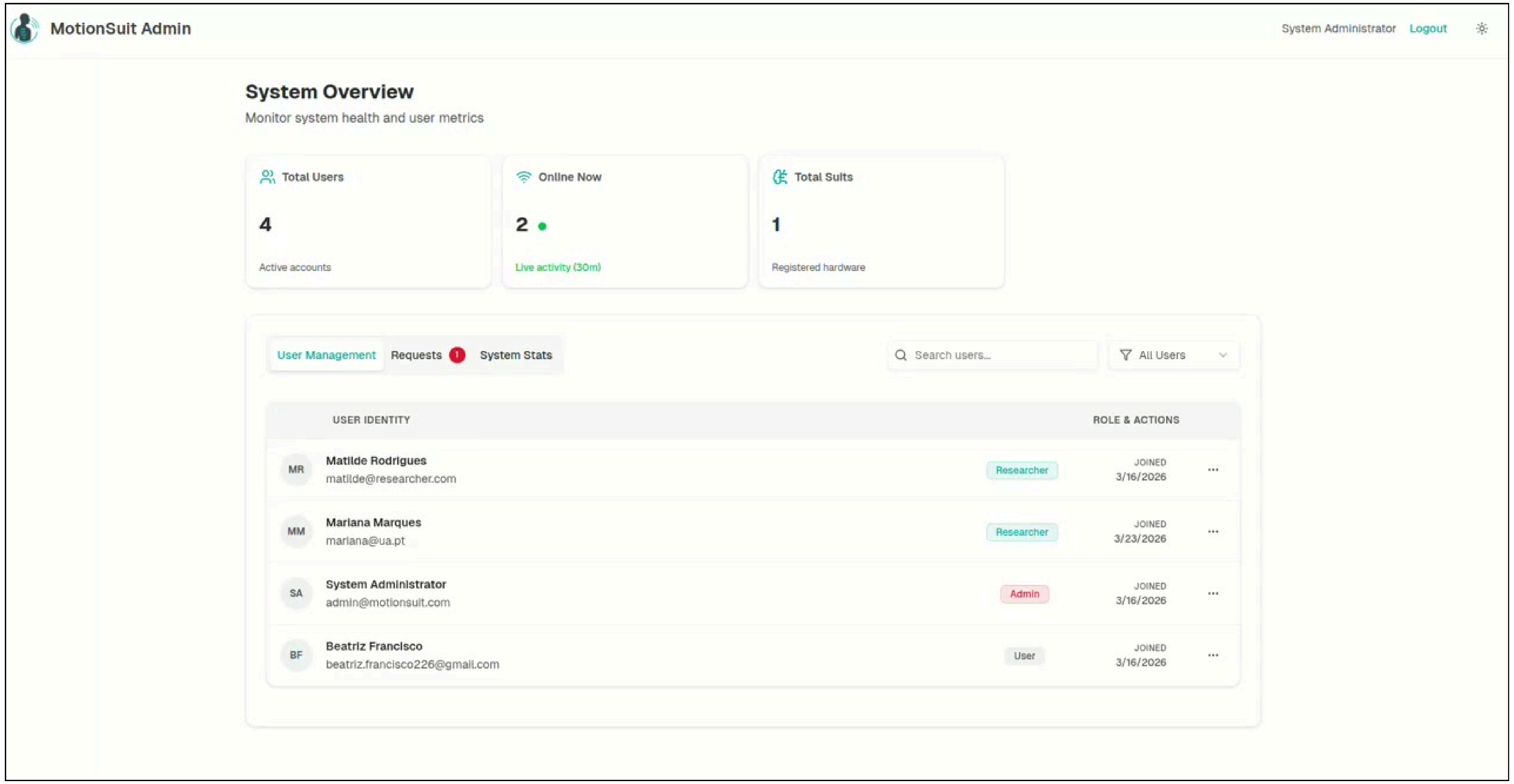


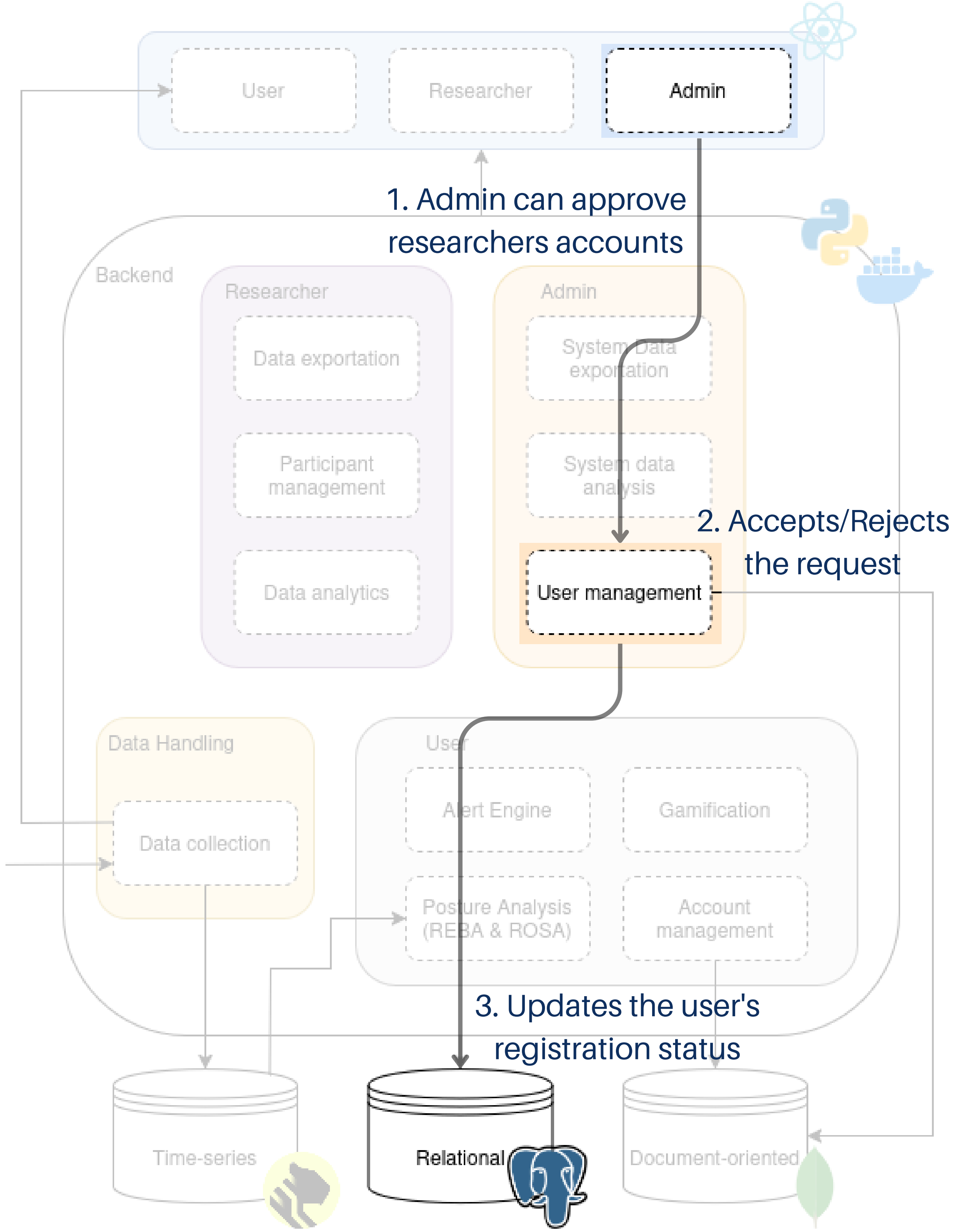
Studies Creation



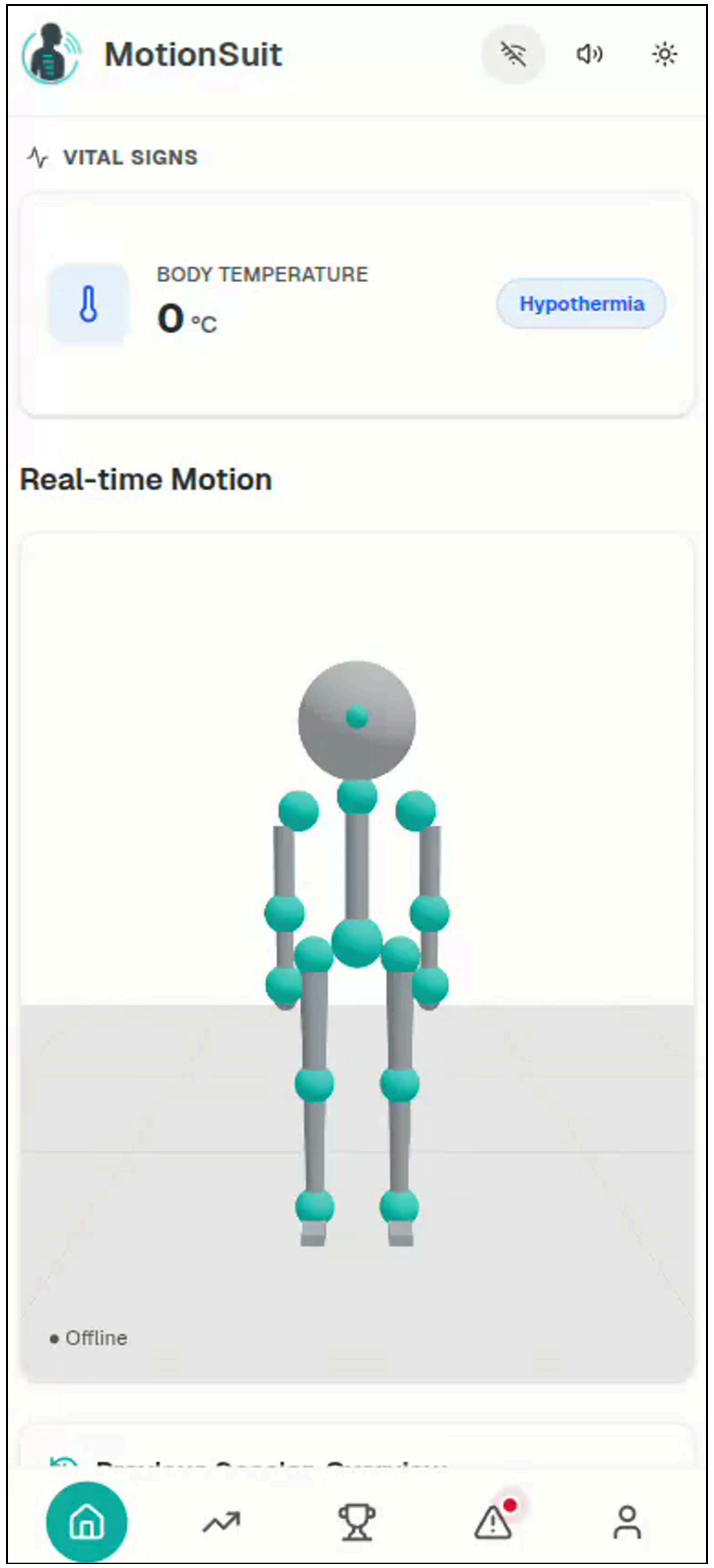


Manage User Accounts





Extra Tasks



Text-to-Speech

CALENDAR

March 24
MS2: Prototype

T-6 — Done
View real-time progress toward goals

T-5 — Done
Set goals to improve posture

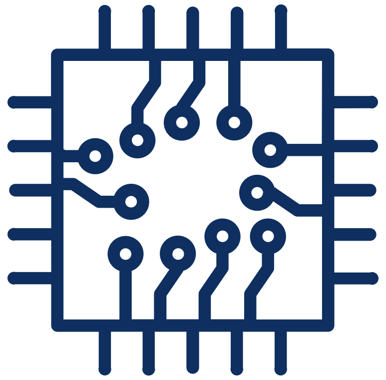
May 5
MS3: Legal requirements, technical and business risks

T-12 — In Review
Compare data from different users

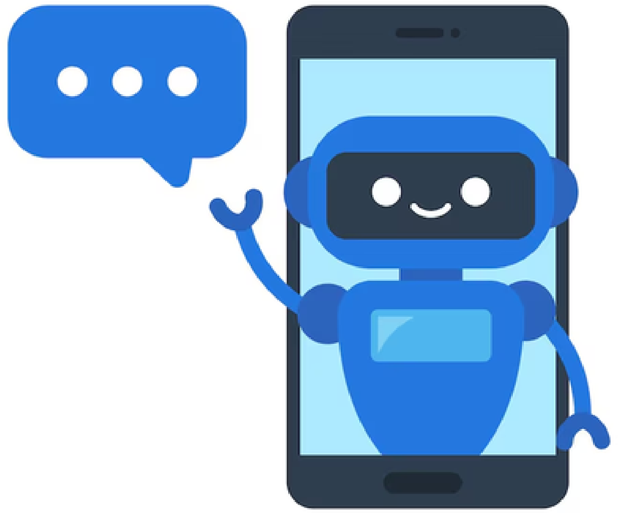
T-3 — In Progress
View real-time metrics on the dashboard

T-7 — To do
Calibrate the suit

NEXT STEPS



Connect all 15 sensors when they arrive



Implement a chatbot to give tips on how users can improve their posture

Connect a smart watch to have access to user heart rate

REFLECTION

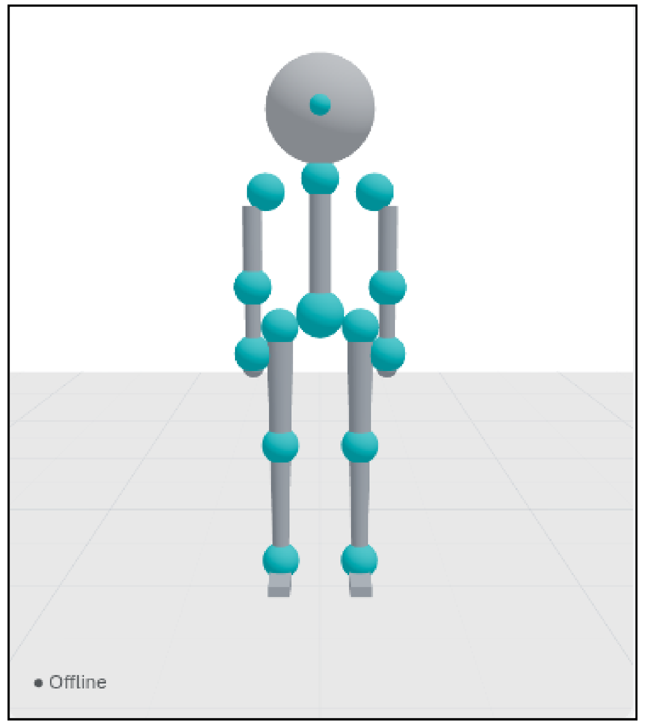
Initial idea

User:
→ Dashboard showing data during session
→ Analyzis page
→ Gamification page
→ Alert system

Admin:
→ Manage users accounts

Researcher:
→ See collected data from all users

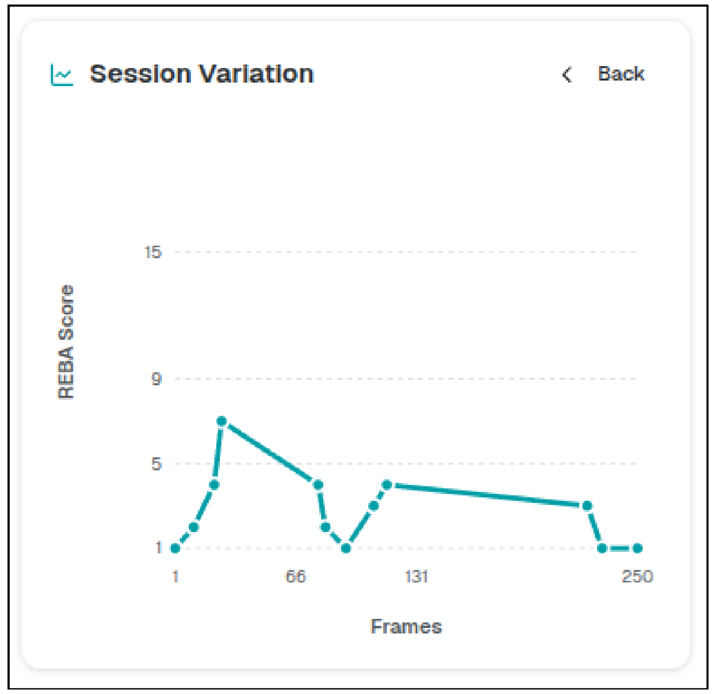
New ideas for the December MVP



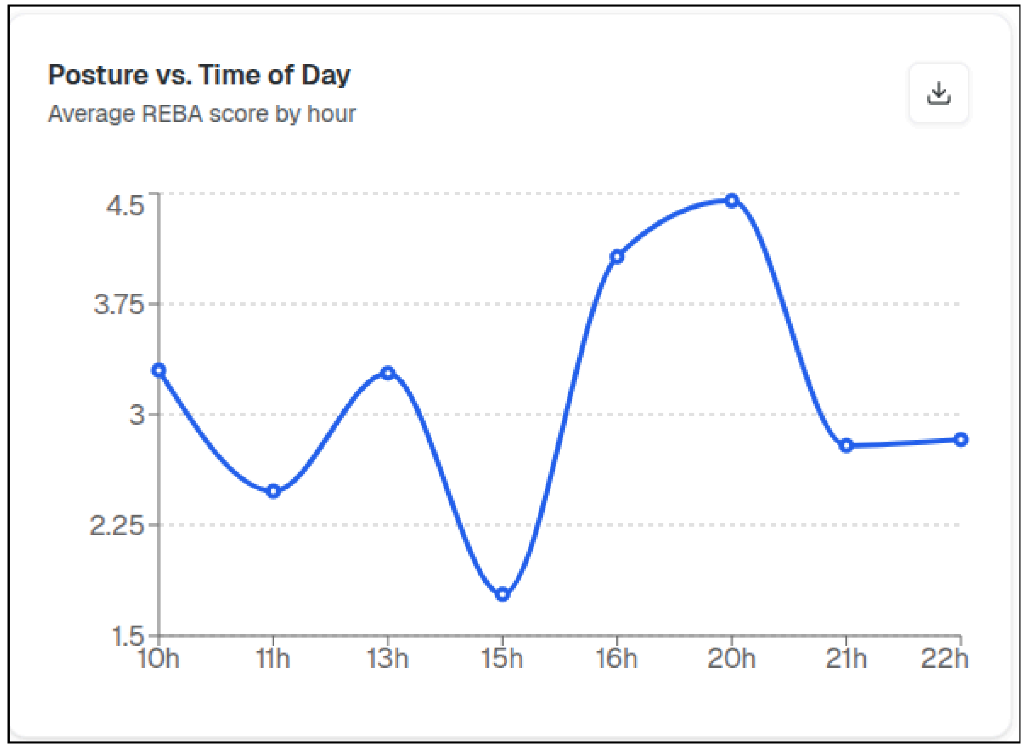
→ Implement a 3D Skeleton

REFLECTION

Ideas for this prototype



→ User access score variation through a graphic



→ Comparative data analysis

→ Text-to-Speech

→ Colorblind Mode

→ Cohort monitoring for research

→Voice-guided postural alerts

THANK YOU!

https://pei-motionsuit.github.io/microsite/

https://pei-motionsuit.github.io/microsite/